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A unified framework for personalized learning pathway recommendation in e-learning contexts
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1 Introduction

With the rapid development of information and communication technology (ICT), e-learning has emerged as an indispensable mode of learning in the educational domain (Lin et al., 2023). By seamlessly integrating education and technology (Al-Fraihat et al., 2020), e-learning presents notable benefits in terms of spatial and temporal flexibility (Zhao et al., 2022), resource availability (Lien et al., 2020), cost reduction (De Freitas et al., 2015), and enhanced control over the learning process (Choudhury & Pattnaik, 2020), among others. In recent years, the proliferation of e-learning systems, particularly the advancement of Massive Open Online Courses (MOOCs), has further expanded learning opportunities for learners (Mangaroska et al., 2021). Nevertheless, these systems frequently adopt the strategy of one-size-

2 Related work

The existing approaches for the PLPRP can be grouped into three main categories: learner-based, knowledge-based, and hybrid recommendation methods, based on the primary factors considered when generating learning pathways for e-learners.

2.1 Learner-based approaches for the PLPRP

When addressing the PLPRP, learner-based approaches prioritize the creation of customized learning pathways that align with the learner’s characteristics and preferences. These methods typically employ data mining techniques to establish the relationship between the learner and the LOs, enabling the selection of personalized

approaches typically employ graphical knowledge representation tools, such as knowledge maps, concept maps, and knowledge graphs, to model the knowledge units and their interdependencies when constructing learning pathways. Durand et al. (2013) have demonstrated that the sequence of LOs is crucial for learners. They introduced graph theory to model LOs and their interdependency relationships as a directed graph and utilized a greedy-based algorithm to find the shortest pathway within this graph-based representation. To support initial and review learning, Pang et al. (2019) proposed a location-based approach that can recommend both prerequisite and subsequent LOs. Building upon the domain knowledge structure, Mu and Yuan (2023) further integrated the individual learner's cognitive state into a cognitive graph-based recommendation approach to construct well-organized learning pathways. More recently, Tzeng et al. (2024) utilized the knowledge map

composition of LOs based on the learners’ specific requirements and contexts. Subsequently, Li et al. ([2023a](#)) employed a comparable approach that initially leveraged a knowledge network to establish the order of knowledge points through sequential links, ultimately obtaining a customized sequence of learning materials aligned with the user’s characteristics.

Hybrid recommendation methods have yielded promising results by combining the advantages of learner-based and knowledge-based approaches. Nevertheless, there is a lack of a unified learning pathway recommendation framework that can effectively and comprehensively incorporate the essential parameters related to learners, LOs, and domain knowledge. From a structural perspective, some studies have not fully integrated the domain knowledge structure with the learner’s cognitive structure, leading to a failure in accurately addressing learners’ intrinsic cognitive

KUs, which are crucial for establishing a logical learning sequence (Zhang et al., 2022a). Figure 1. provides an example illustrating the prerequisite relationships between three KUs.

Definition 2: Learner’s current cognitive graph The learner’s current cognitive graph is represented as $C(i) = (K(i), E(i))$, where $K(i)$ denotes the set of KUs that the learner u_i has effectively grasped and $E(i)$ denotes the relationships between these KUs. A learner’s level of mastery for each specific KU can be determined by integrating objective domain knowledge tests and subjective self-assessment (Xie et al., 2017). When the resulting value exceeds the predefined threshold, it indicates that the learner has effectively mastered the corresponding KU.

Definition 3: Learning objective The learning objective, denoted as O_i , refers to the set of the targeted KUs that the learner u_i expects to acquire. It can be derived through analyzing the learner’s input query sentence using natural language processing methods (Shi et al., 2020).

Definition 4: Learner’s goal knowledge graph The learner’s goal knowledge graph is represented as $G(i) = (K(i), E(i))$ which encompasses the essential KUs and their relationships that the learner u_i needs to acquire in order to successfully fulfill their learning objectives. It is derived by mapping the domain knowledge graph G with the learner’s current cognitive graph $C(i)$ and their specific learning objectives O_i . The knowledge graph supports the learner’s transition from their current cognitive state to the desired target state.

Definition 5: Learning object A LO, denoted as LO_i , refers to any reusable digital resource that facilitates learning (Wiley, 2000). Each LO may contain one or more

Fig. 1 An example of prerequisite relationships between three KUs

KUs, where L_i represents the set of KUs within in the L_i . The size of L_i , denoted as $N_{L_i} = |K_{L_i}|$, indicates the level of granularity of L_i (Balatsoukas et al., 2008). A larger value of N_{L_i} implies a coarser granularity of L_i , while a smaller value indicates a finer granularity. Online learning environments typically contain diverse types of LOs, such as atomic LOs that cover a single specific KU, modular LOs comprising a series of interconnected KUs, and large-sized LOs corresponding to a complete lesson or an entire course (Nabizadeh et al., 2020b). Figure 2 presents examples of different types of LOs.

Definition 6: Learning Pathway A learning pathway, denoted as P_i , represents a tailored sequence of LOs designed for the learner u_i , which can effectively assist the learner in achieving the established learning objective. The length of feasible learning pathways for a specific learner can vary due to the different levels of granularity among the various types of LOs.

3.2 Problem statement

Suppose there is an extensive collection of LOs within an online learning system, represented as $L = \{L_1, L_2, \dots, L_n\}$. The task of the PLPRP is to select M appropriate LOs from L and arrange them into a suitable learning pathway P_i for each individual e-learner u_i . Figure 3 provides a specific example illustrating the PLPRP. It can be observed from Fig. 3 that there are multiple feasible learning pathways available for a given learner. Thus, the main objective of the PLPRP is to identify

Fig. 2 Examples of different types of LOs

Fig. 3 Example of the PLPRP

the best learning pathway that can effectively guide the learner towards their learning goals while maximizing their learning outcomes.

3.3 Problem modeling

To comprehensively formulate the problem requirements of PLPRP, we introduce a novel two-hierarchy modeling architecture that initially modeling this problem from structure and content hierarchies respectively and then integrate them into a unified mathematical model. The modeling architecture of the PLPRP is depicted in Fig. 4.

3.3.1 Modeling from the structure hierarchy

From the perspective of structure, we should build coherent and efficient learning pathways for learners to meet their cognitive learning requirements and maximize their learning efficiency. To achieve it, several constraints and functions are defined. It is crucial to highlight that we formulate the learning pathway structure at the KU level rather than the LO level since it is challenging to directly establish the prerequisite relationships between LOs of varying levels of granularity, as demonstrated in Fig. 3.

- (1) Cognitive order constraint.

Fig. 4 The two-hierarchy modeling architecture of the PLPRP

To provide learners with a cohesive learning experience, it is essential to ensure that all KUs within the final learning pathway are satisfied their prerequisite relationships. This arrangement helps prevent learners from experiencing confusion or a lack of direction during their learning journey. The cognitive order constraint is defined as follows.

$$[k_{p_i}^m, k_{p_i}^n] \in E(i) \quad \forall k_{p_i}^m, k_{p_i}^n \in K_{p_i}^\diamond \tag{1}$$

where $E(i)$ represents the set of prerequisite relationships among the KUs in $G(i)$. $K_{p_i}^\diamond = \bigcup_{j=1}^L K_{L_j} (L_j \in P_i)$ represents the set of unduplicated KUs within P_i , in which the order of these KUs corresponds to their order in P_i . $k_{p_i}^m \in K_{p_i}^\diamond$ and $k_{p_i}^n \in K_{p_i}^\diamond$ denote any two KUs in $K_{p_i}^\diamond$.

(2) Cognitive goal constraint.

To ensure the learner’s successful attainment of the learning objectives, it is necessary that the final learning pathway incorporates all the essential KUs contained in

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the learner’s goal knowledge graph G_i). The cognitive goal constraint is expressed as follows.	
$K_i \subseteq K_{p_i}$	(2)
where K_i denotes the set of essential KUs in G_i that the learner u_i must acquire to support the achievement of the learning objectives.	
(3) Learning coherence function	
We define the learning coherence function, denoted as $C(P_i)$, which quantifies the coherence of the learning pathway P_i by considering the prerequisite relationships between KUs. The learning coherence function is defined as follows:	

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e-learners. Personalization parameters play a significant role in guiding the selection of appropriate LOs (Nabizadeh et al., 2020b). Previous research has highlighted that several factors need to be considered when tailoring the content of these pathways, including the e-learner’s competency level, available time, media preference, and other factors such as their requirements and preferences for novelty, authority, popularity, and evaluation (Shi et al., 2020; Wan & Niu, 2016; Zhu et al., 2018). Thus, this paper proposes these seven dimensions as the personalization parameters for optimizing the content of the learning pathways.	
(1) Competency level.	
According to the cognitive load theory, learning content that is excessively dif-	

The degree of suitability between the learner's available learning time and the total suggested study time of all the recommended LOs, denoted as $2 \text{ } P_i \text{ } 3$ is evaluated using the following formula:

$$S_2 \begin{bmatrix} P_i & * \end{bmatrix} \begin{cases} \sum_{at_i}^M st_j \text{ ?}_i, & \text{if } \sum_{j=1}^M st_j \text{ ?}_i \geq 0 \\ 0, & \text{otherwise} \end{cases} \quad (8)$$

where t_j represents the suggested study time of j , which can be obtained by taking into account teachers' instructional experience and the mean learning time of previous learners (Wan & Niu, 2018). at_i represents the available time of learner u_i .

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(4) Novelty preference.
Studies have shown that learners have personalized inclinations towards the novelty of learning resources (Bulathwela et al., 2020). Some learners exhibit a heightened motivation to acquire recently introduced learning materials, as they provide the opportunity to access the most current and cutting-edge information and knowledge. Meeting learners’ demand for novelty in resources can enhance their concentration, enthusiasm, and active engagement throughout the learning process (Lowe et al., 2010). The degree of suitability between the novelty levels of the recommended LOs and learner’s preferences for novelty, denoted as $S_4[P_i]$, can be determined based on the release time.

in a specific domain. On the other hand, a_j represents the institutional authority level of the publisher for L_j , which can be assessed by considering the ranking and influence of the publisher. Both a_j and w_j are normalized into the same interval, and w_{ha} and w_{hp} are the respective weights assigned to them. $ap_i \in \{0,1\}$ denotes whether the learner prioritizes the authority of resources. $Suit_A[u_i, L_j]$ indicates the degree of suitability between the authority level of u_i and the authority preference of L_j .

(6) Popularity preference.

Popularity is an important factor that reflects the frequency with which resources are utilized by previous learners and can significantly influence the choices made by

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solution construction strategy to flexibly generate feasible learning pathways that satisfy the constraints of the PLPRP. To further improve the optimization performance for addressing the PLPRP, we also redefine several important components such as the pheromone initialization method and the local and global pheromone update rules. The flowchart of the proposed MACO is presented in Fig. 5.
4.1 Candidate LO identification
In online learning systems, the abundance of available LOs often results in a significantly large search space. To enhance the efficiency of the recommendation process, we introduce a candidate LO identification algorithm that filters out irrelevant LOs and selects only relevant ones, thereby reducing the size of the search space. Specifi-

Fig. 5 The flowchart of the proposed MACO

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($Rep \in [L_r, L_s]$). The weight parameter w is utilized to adjust the relative importance of these two aspects. A higher value of $\alpha(r, s)$ indicates a higher a greater likelihood of incorporating L_s into the current learning pathway.

4.2.2 Initialization of the pheromone concentration matrix

The initialization setting of the pheromone concentration directly influences the search ability in the early phase and can further influence the algorithm’s convergence speed (Paniri et al., 2021). In the original ACS algorithms, the initial values of pheromone trails between each pair of nodes are set to a single constant value. In line with the work of Jovanovic et al (2019), we adopt a greedy method for initializing the pheromone matrix to enhance the algorithm’s search process. The initial

feasible neighbors, which are candidate LO nodes satisfying the given constraints, to construct a feasible set of candidates for the ant. As depicted in Algorithm 1, we first construct a feasible set of KUs, denoted as $KU_k(i, t)$, by selecting nodes from $G_k(i, t)$ that do not have any prior nodes. Next, we consider the corresponding LOs covering these KU nodes and only choose those that not conflict with the remaining unvisited KUs as feasible neighbors. These selected LOs constitute the allowable set of LOs, denoted as $\Omega_k(i, t)$, for each ant k . By incorporating this dynamic neighbor selection mechanism, we ensure that the generated learning pathways adhere to the cognitive order constraints. Furthermore, this approach significantly reduces the number of feasible neighbors compared to considering all unvisited LOs. This reduction in candidate neighbors enhances the algorithm’s search capability and improves its efficiency.

Algorithm 1 Dynamic neighbor selection

4.3.2 Heuristic strategy for the starting point assignment

In traditional ACO algorithms, the path construction phase typically begins with randomly generated positions, without utilizing any heuristic information (Deng et al., 2022). However, for the considered PLPRP, we design a heuristic strategy for the starting point assignment that uses the suitability degree as the heuristic information

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to determine the selection probability and then choose the starting points for each ant by the roulette rule. The probability of selecting the node r as the starting point for ant k , denoted as P_r^k , is calculated as follows.

$$P_r^k = \frac{\text{Suit}(\bar{u}_i, L_r)}{\sum_{r \in \Omega_k(i, t)} \text{Suit}(\bar{u}_i, L_r)} \quad \forall r \in \Omega_k(i, t) \tag{26}$$

where $\Omega_k(i, t)$ indicates the set of feasible starting nodes for ant k , as determined by Algorithm 1. This strategy facilitates the algorithm’s exploration towards the target regions, thus decreasing blind search in the early stages and promoting algorithm convergence.

4.3.3 State transition rule for the next node selection

contained within the currently selected LO node l_j must be deleted from $G(i, t)$ in order to obtain the updated $G(i, t + 1)$, while any LO nodes that fully cover the KUs in l_j must be eliminated from $L(i, t)$ to derive the updated $L(i, t + 1)$. This update operation ensures the fulfillment of the uniqueness constraint, whereby each LO node can only be selected once during the construction of the learning pathway solution.

To meet the cognitive goal constraint, we adopt the condition that all nodes in $G(i)$ must be fully visited as the termination criterion for solution construction. When there are no remaining nodes in $G(i)$, it indicates the completion of a full learning pathway solution. This solution encompasses all the necessary KUs that support the learner’s transition from their current cognitive state to the desired target cognitive state.

4.3.5 Illustrative example for the learning pathway solution construction

Figure 6 provides an illustrative example of the proposed knowledge graph-based solution construction strategy. In this example, the learner’s goal knowledge graph indicates that only k_1 does not have a prior node. Thus, k_1 is initially selected into the feasible set of KUs, while l_1 and l_2 form the corresponding allowable set of LOs. By applying Eq. (26), l_2 is chosen as the starting point for the learning pathway. Subsequently, the learner’s goal knowledge graph and the LO set are dynamically updated. k_3 and k_4 are added to the feasible KU set, and l_4 , l_5 , and l_6 are included in the allowable LO set. Employing the pseudorandom proportional rule, l_6 is selected as the second node in constructing the learning pathway. After this selection, the learner’s goal knowledge graph and the LO set are updated once again. At this stage, only k_6 remains in the feasible KU set, and its corresponding LOs, l_7 and l_8 are the remaining candidate LOs. Based on the pseudorandom proportional rule, l_8 is chosen to continue building the learning pathway. Through the update operation, the learner’s goal knowledge graph becomes null, indicating the completion of the learning pathway. In this example, the identified feasible learning pathway is $l_2 \rightarrow l_6 \rightarrow l_8$.

Fig. 6 An illustrative example of the proposed knowledge graph-based solution construction strategy

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4.4 Local pheromone update rule
During the iterative search process, after one ant finishes the construction of a learning pathway solution, the local pheromone update rule is immediately performed based on Eqs. (29) and (30). Following this rule, the pheromone levels on all the arcs that the ant has traversed will be reduced, which can help to promote the exploration ability for unknown regions and enhance the solution diversity (Huang et al., 2020).

$$\tau(r,s) \leftarrow (1-\rho) \cdot \tau(r,s) + \rho \cdot \Delta \tau(r,s) \tag{29}$$

$$\Delta \tau(r,s) = \frac{1}{L_{\text{ant}}}, \text{ if } (r,s) \in S$$

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5 Computational experiment and result analysis
This section presents computational experiments conducted to examine the performance of the proposed MACO algorithm from a computational perspective.
5.1 Test datasets
According to our knowledge, there is currently no publicly available and standardized dataset specifically designed for the PLPRP. Hence, we generated a set of test datasets with various scales and complexities for this study. The detailed statistics of these test datasets are presented in Table 1, where #KU, #LO, #Type, and #Para represent the numbers of KUs, LOs, the types of LOs, and the number of personali-

for previous studies to simulate stochastic learning behaviors (Bian et al., 2019; Li et al., 2023a). IA and GA are well-known meta-heuristic algorithms with good performance in solving optimization problems (Zhang et al., 2022b; D’Angelo & Palmieri, 2021). ACS (Schyns, 2015) is the foundational framework for the proposed MACO, enabling the evaluation of performance enhancements resulting from the modifications. IGA is a state-of-the-art method for the PLPRP, using a two-stage mechanism to generate a feasible concept sequence and then identify the optimal sequence of LOs (Benmesbah et al., 2023).

5.3 Parameter configurations

The proposed MACO algorithm requires several key parameters to be set, as they

The number of teaching resources used in the classroom for the purpose of teaching the subject of the course	$n_{\mathcal{E}M}$	$\frac{5}{310}$	$\frac{7}{70}$	$\frac{6}{410}$	$\frac{8}{410}$	$\frac{2}{410}$	$\frac{9}{410}$	$\frac{7}{70}$	$\frac{6}{800}$	$\frac{8}{800}$	$\frac{9}{310}$	$\frac{1}{800}$	$\frac{6}{800}$
	$O_{CA}M$	$\frac{5}{310}$	$\frac{7}{70}$	$\frac{6}{410}$	$\frac{8}{410}$	$\frac{2}{410}$	$\frac{7}{70}$	$\frac{9}{410}$	$\frac{3}{800}$	$\frac{5}{800}$	$\frac{9}{310}$	$\frac{4}{800}$	$\frac{2}{410}$
	$n_{\mathcal{E}M}$	$\frac{8}{310}$	$\frac{5}{410}$	$\frac{9}{810}$	$\frac{2}{310}$	$\frac{8}{310}$	$\frac{2}{810}$	$\frac{9}{610}$	$\frac{4}{610}$	$\frac{3}{310}$	$\frac{2}{310}$	$\frac{9}{810}$	$\frac{8}{810}$
	S_{CA}	$\frac{5}{310}$	$\frac{1}{70}$	$\frac{6}{410}$	$\frac{6}{810}$	$\frac{0}{410}$	$\frac{9}{910}$	$\frac{8}{810}$	$\frac{1}{510}$	$\frac{8}{310}$	$\frac{9}{310}$	$\frac{8}{810}$	$\frac{1}{310}$
	$n_{\mathcal{E}M}$	$\frac{5}{310}$	$\frac{3}{70}$	$\frac{4}{410}$	$\frac{7}{410}$	$\frac{2}{410}$	$\frac{6}{710}$	$\frac{8}{710}$	$\frac{4}{710}$	$\frac{8}{410}$	$\frac{4}{310}$	$\frac{5}{810}$	$\frac{5}{410}$
	A_F	$\frac{5}{310}$	$\frac{7}{70}$	$\frac{6}{410}$	$\frac{8}{410}$	$\frac{7}{610}$	$\frac{0}{310}$	$\frac{8}{310}$	$\frac{7}{310}$	$\frac{7}{910}$	$\frac{0}{710}$	$\frac{9}{410}$	$\frac{8}{210}$
	$n_{\mathcal{E}M}$	$\frac{4}{310}$	$\frac{3}{810}$	$\frac{7}{410}$	$\frac{6}{810}$	$\frac{7}{310}$	$\frac{9}{410}$	$\frac{2}{310}$	$\frac{6}{410}$	$\frac{0}{610}$	$\frac{7}{410}$	$\frac{9}{310}$	$\frac{7}{410}$
	A_G	$\frac{8}{510}$	$\frac{5}{310}$	$\frac{2}{410}$	$\frac{1}{210}$	$\frac{6}{610}$	$\frac{4}{810}$	$\frac{3}{310}$	$\frac{4}{310}$	$\frac{2}{310}$	$\frac{5}{310}$	$\frac{2}{410}$	$\frac{7}{410}$
	$n_{\mathcal{E}M}$	$\frac{7}{410}$	$\frac{5}{910}$	$\frac{8}{410}$	$\frac{9}{310}$	$\frac{6}{610}$	$\frac{6}{310}$	$\frac{8}{610}$	$\frac{4}{310}$	$\frac{2}{310}$	$\frac{9}{410}$	$\frac{3}{810}$	$\frac{9}{410}$
	A	$\frac{0}{210}$	$\frac{5}{710}$	$\frac{6}{410}$	$\frac{8}{410}$	$\frac{7}{210}$	$\frac{5}{410}$	$\frac{5}{610}$	$\frac{7}{810}$	$\frac{2}{410}$	$\frac{9}{310}$	$\frac{7}{610}$	$\frac{8}{810}$
	$n_{\mathcal{E}M}$	$\frac{7}{810}$	$\frac{7}{810}$	$\frac{9}{810}$	$\frac{5}{310}$	$\frac{1}{510}$	$\frac{9}{610}$	$\frac{1}{410}$	$\frac{4}{510}$	$\frac{6}{610}$	$\frac{4}{510}$	$\frac{4}{510}$	$\frac{1}{410}$
	S_R	$\frac{0}{310}$	$\frac{7}{610}$	$\frac{7}{310}$	$\frac{6}{310}$	$\frac{1}{410}$	$\frac{8}{310}$	$\frac{5}{410}$	$\frac{3}{810}$	$\frac{9}{410}$	$\frac{1}{410}$	$\frac{8}{810}$	$\frac{1}{610}$

$t_{\mathcal{E}D}$	1	2	3	4	5	6	7	8	9	9	1	1
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(2) Analysis of learning suitability.

Learning suitability assesses an algorithm’s ability to personalize learning pathways according to individual requirements. Among the considered algorithms, it is obvious that RS performs the weakest in terms of personalizing learning pathways across all cases, indicating challenges in identifying suitable learning resources. Conversely, IGA, ACS, and MACO outperform GA and IA in learning suitability by incorporating Learner-to-LO matching. While all three algorithms show comparable personalization performance in Scenario 1, MACO consistently outperforms the others. Remarkably, even in the highly complex Scenario 3, MACO displays superior personalization capabilities, highlighting the effectiveness of our designed heuristic information function in discovering suitable learning content for learners. To sum-

complexity increase, the performance of IGA gradually becomes inferior to that of MACO. Clearly, MACO exhibits the strongest ability to minimize KU repetition with learning pathways, indicating that incorporating redundancy between LOs in the heuristic information function motivates ants to choose LOs with less repetition, thereby ensuring pathway efficiency. As a result, MACO demonstrates remarkable performance in enhancing the efficiency of learning pathways. This can effectively reduce learners' cognitive load and allow them to focus on essential knowledge aligned with their learning objectives, leading to improved learning outcomes and overall satisfaction.

5.4.2 Analysis of stability performance

5.4.3 Analysis of utilization rate

The utilization rate, defined as the proportion of recommended LOs to the total number of candidate LOs, was employed to analyze the algorithms’ LO utilization capabilities (Wan & Niu, 2016). A lower utilization rate indicates stronger algorithm utilization. All algorithms except MACO adopted a two-stage planning strategy, similar to IGA, to ensure the learning pathways that align with learners’ cognitive needs. As a result, their utilization rates were equal to the ratio of the number of KUs in the learner’s target subgraph to the total number of candidate LOs, i.e., 0.3333 in this study, serving as the baseline for this metric. The utilization rate results are summarized in Table 3.

According to the results in Table 3, it can be observed that MACO and the

	2	1
	1	0
	9	0
	9	0
	8	0
	7	0
Research methodology of the OCCAM network configuration for the performance analysis	6	0
	5	0
	4	0
	3	0
	2	0
	1	0
3 r b t	te g D	h o n e U

Before the experiment began, a comprehensive explanation of the experimental procedure was provided to all participants. Furthermore, a well-designed questionnaire was distributed to the students in the Recommended group to gather data on their learning characteristics and preferences. The collected raw data was then processed and analyzed, and our proposed method was implemented to determine the optimal learning pathways for these students. Throughout the experiment, all students used the same online learning platform, with the only distinction being the presentation and order of the learning content. An online video recording tool was used to record each student’s learning process, including the LOs they visited, the duration of their learning sessions, and the order in which they accessed the LOs. The overall learning time was determined by calculating the total duration from the start to the end of each recorded session.

recommended learning pathways. This suggests that our proposed method can effectively align with learners’ cognitive processes by connecting KUs based on their dependency relationships. These finding resonates with previous research, such as Zhu et al. (2018) and Shi et al. (2020), who emphasized the importance of knowledge dependency in organizing systematic and coherent learning pathway. In contrast, the Self-organized group exhibited longer learning pathway lengths, indicating a higher occurrence of repetitive learning behaviors among its students. This highlights the challenges that many students face when attempting to determine the sequential learning order of different KUs without guidance. Without accurate preconceptions about the relationships between these units, students may struggle to integrate them into a systematic knowledge framework. As a result, unnecessary repetition of learning becomes more prevalent as students strive to comprehend and

Learning satisfaction can reflect students' subjective perceptions regarding their learning experience. The findings obtained from the study revealed a significant difference in satisfaction levels between the Recommended group and the Self-organized group (p -value < 0.01). This suggests that the majority of students in the Recommended group exhibit a high level of recognition and acceptance towards the provided recommendations, which aligns with the findings of Wan and Niu (2018). Guided by a well-established learning pathway, students in the Recommended group can better organize and regulate their learning process, leading to effective knowl-

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Data availability The datasets generated and/or analysed during the current study are available from the first author, Yaqian Zheng, upon reasonable request.

Declarations

Conflict of interest None.

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